

Financial Frictions, Occupational Choice and Economic Inequality

Lian Allub, Andrés Erosa

PII: S0304-3932(18)30620-2  
DOI: <https://doi.org/10.1016/j.jmoneco.2018.12.003>  
Reference: MONEC 3053

To appear in: *Journal of Monetary Economics*

Received date: 25 April 2016  
Revised date: 12 December 2018  
Accepted date: 17 December 2018

Please cite this article as: Lian Allub, Andrés Erosa, Financial Frictions, Occupational Choice and Economic Inequality, *Journal of Monetary Economics* (2018), doi: <https://doi.org/10.1016/j.jmoneco.2018.12.003>



This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

**Highlights**

- Cross-country differences in entrepreneurship are explained by self-employment differences
- Develop a model that distinguishes between self-employed entrepreneurs and employers
- Financial Frictions affect the average managerial and working ability of the economy
- Modelling heterogeneity in two skills amplifies the impact of financial frictions on output here the sovereign chooses debt maturity rationalizes these stylized facts.

# Financial Frictions, Occupational Choice and Economic Inequality<sup>☆</sup>

Lian Allub<sup>a</sup>, Andrés Erosa<sup>b,\*</sup>

<sup>a</sup>CAF-Development Bank of Latin America, Madero 900, Piso 14, Torres Catalinas Plaza, Ciudad Autónoma de Buenos Aires, Argentina, C1106

<sup>b</sup>Calle Madrid 126, Universidad Carlos III de Madrid, Getafe, Spain, 28903

## Abstract

The Lucas (1978) model is extended to incorporate heterogeneity in working ability and a time allocation decision by entrepreneurs (work versus manage). Financial frictions distort not only the average skill of entrepreneurs but also the average skill of workers. The model economy accounts for half of the association between entrepreneurship and external finance to GDP in the data, whereas a standard span of control model explains only about one tenth. The variation in entrepreneurship is mostly due to the variation in self-employed entrepreneurs rather than in employers. Moreover, financial frictions have larger effects on output per worker, TFP, and inequality.

**Keywords:** Financial frictions, entrepreneurs, employers, self-employed, TFP, inequality

## 1. Introduction

A theory of entrepreneurship that distinguishes between entrepreneurs that are self-employed and those that are employers is built in order to study the impact of financial frictions on occupational sorting, output, and inequality. The importance of modelling this distinction is motivated by cross-country evidence on the negative association between entrepreneurship and the ratio of external finance to GDP across countries (see Figure 1). While the fraction of employers tends to be slightly increasing with financial development, the fraction of self-employed households is strongly negatively associated with external finance to GDP. As a result, the evidence suggests that the variation in self-employed entrepreneurs is crucial for understanding the negative relationship between entrepreneurship and financial development in the cross-country data.

Evidence from Brazil on household earnings inequality between and within occupations points that the mean earnings of entrepreneurs is not higher than that of workers.<sup>1</sup> Moreover, Figure 2 shows that there is substantial earnings heterogeneity within all occupations (workers, employers, and self-employed) so that there are households at the top and bottom of the income distribution in all occupations. The standard span of control model with heterogeneity in one skill cannot account for these observations.

The standard Lucas (1978) framework is extended by assuming that agents are heterogeneous in managerial and working skills, as in Jovanovic (1994), and by modeling a time allocation decision in which entrepreneurs divide their time between managing their businesses and working. Households choose how

<sup>☆</sup>We thank our discussant Francisco Buera and seminar participants at CAF "Taller RED 2013" in Buenos Aires for helpful and insightful comments. We also thank Inés Berniell, Luisa Fuster, Ramon Marimon, the associate editor Ben Moll and one anonymous referee for valuable comments and suggestions. Authors acknowledge financial support from CAF-Development Bank of Latin America. Erosa acknowledges financial support from the Ministerio de Economía y Competitividad of Spain (grants ECO2015-68615-P and MDM 2014-0431) and Comunidad de Madrid, MadEco-CM (S2015/HUM-3444). Allub acknowledges financial support from the Max Weber Program at EUI. All errors remain ours.

\*Corresponding author

Email addresses: lallub@caf.com (Lian Allub), andres.erosa@uc3m.es (Andrés Erosa)

<sup>1</sup>In Brazil, mean earnings are lower for self-employed entrepreneurs than for workers, and are higher for employers than for workers. These facts apply to other countries as well. See Hamilton (2000), Hurst and Pugsley (2011), Poschke (2013), Vissing-Jorgensen and Moskowitz (2002) for papers documenting facts on entrepreneurial returns in the US.

much to consume, save and whether to be a worker or an entrepreneur (self-employed or an employer). Entrepreneurs decide how much time to devote to the supply of labor and/or managerial inputs and how much (outside) labor to hire. Financial frictions are modeled with a collateral constraint limiting the amount of capital that entrepreneurs can use, as in Buera and Shin (2013) and Moll (2014).

The theory implies that some entrepreneurs will rely only on their own labor and managerial inputs and become self-employed, while other entrepreneurs will decide to hire labor services in the market and become employers. By modeling heterogeneity in the form of two skills the theory can distinguish between a comparative advantage in entrepreneurship (a high ratio of managerial to working skills) and an absolute advantage (a high value of both skills). Heterogeneity in absolute advantage implies that both at the top and bottom of the income distribution there are entrepreneurs and workers, which is needed for the theory to be consistent with household evidence on earnings inequality between and within occupations.

The model economy is calibrated to Brazilian household and aggregate data. The skill distribution and its evolution over time is chosen to match moments on occupational choices, occupational transitions, and earnings inequality between and within occupations. To benchmark the results, we also calibrate a version of the model economy in which households only differ in their managerial skills and entrepreneurs do not face a time allocation problem, as in the Lucas (1978) model.

One novel implication of the theory in this paper is that financial frictions distort not only the productivity of entrepreneurs but also the productivity of workers. In the baseline economy, as financial frictions are reduced the changes in the equilibrium sorting of workers are such that the productivity of employers increases and that of workers decreases. The first effect increases labor demand and the second one decreases labor supply in *effective units* of labor. This asymmetric effect is associated to a large increase in the number (quantity) of workers in order to equate labor supply and labor demand, which works through a large reduction in the fraction of self-employed households. As a result, the elimination of financial frictions causes the rate of entrepreneurship to decrease much more in the baseline economy (from 32% to 12%) than in the standard model economy (from 10.5% to 7%).

The quantitative performance of the theory is assessed by computing, for the two calibrated model economies, equilibrium allocations that differ on financial frictions. We find that the baseline model accounts for half of the association in the cross-country data between entrepreneurship and external finance to GDP while the standard model explains only about one tenth. The baseline model also explains the fact that the fraction of employers shows a mildly positive association with the external finance to GDP ratio in the cross-country data and that the variation in entrepreneurship is mostly due to the variation in self-employment rates. In the baseline economy, a reduction in financial frictions shifts households from self-employment into paid work, thereby increasing the share of employee compensation in national income. This prediction of the theory is consistent with the cross-country data. Quantitatively, the baseline economy accounts for about 30% of the empirical association between the ratio of employee compensation to GDP and the ratio of external finance to GDP. The standard model cannot account for this evidence because it abstracts from self-employment.

The output gain of eliminating financial frictions in the baseline economy is much larger than the one in the standard model economy (58% versus 47%). Crucial to this result is that the decrease in the fraction of entrepreneurs is more than 5 times bigger in the baseline economy than in the standard model economy (0.20 versus 0.035), implying that the reallocation of capital and labor towards the most productive entrepreneurs is more important in the baseline economy than in the standard model. The larger reallocation of capital towards a smaller number of entrepreneurs implies a larger credit expansion in the baseline economy, leading to a higher increase in output.

Given that the baseline economy provides a good account of household inequality in Brazil, the impact of financial frictions on household inequality is evaluated. The elimination of financial frictions in the baseline model economy leads to a substantial reduction in the inequality across households, decreases the persistence of income over time and the correlation of earnings and assets. Interestingly, the most important effect of eliminating financial frictions on inequality is not on the reduction of inequality *per se* but on the sources that drive inequality. The elimination of financial frictions allows talented households to obtain the full reward to their skills increasing the proportion of the income variance explained by skills.

The paper relates to a large literature in macroeconomics investigating the consequences of financial

frictions on aggregate output, entrepreneurship, and productivity.<sup>2</sup> Relative to this literature, we show that self-employment and heterogeneity in working skills are important for understanding the strong negative association between entrepreneurship and financial development in the cross-country data as well as the output and productivity gains of removing financial frictions.

Jovanovic (1994) extends the Lucas model with two dimensional skill heterogeneity to show that entrepreneurs are not necessarily drawn from the top of the ability distribution. Our paper extends the analysis in Jovanovic by considering a dynamic model with financial frictions and a time allocation problem of entrepreneurs. D’Erasmus and Boedo (2012), Hurst and Pugsley (2011), Poschke (2013), Yurdagul (2017) develop theories of entrepreneurship with small businesses.<sup>3</sup> None of these papers account for own-account workers.

## 2. The model

Consider a closed economy that is in steady state and is populated by a large number (normalized to 1) of infinitely-lived households. Households are heterogeneous in wealth and in working ( $z_w$ ) and managerial skills ( $z_m$ ). They face uninsurable idiosyncratic shocks to their skills and maximize expected discounted utility over stochastic consumption sequences.

### 2.1. Production technology

Following Lucas (1978), output is produced with a constant returns to scale production technology in managerial, labor, and capital inputs. Entrepreneurs can only use their managerial input because there is no market for managers. The supply of the managerial input is equal to the product of the households’ managerial ability ( $z_m$ ) and the time devoted to managing a business ( $t_m$ ). The output produced by a household supplying  $m = z_m t_m$  units of managerial input and using  $k$  units of capital and  $n$  efficiency units of labor is:

$$Y(m, k, n) = m^\gamma k^\eta n^\theta, \text{ where } \gamma + \eta + \theta = 1. \quad (1)$$

The time allocation decision of entrepreneurs ( $t_m \in [0, 1]$ ) is modeled to introduce self-employment (own-account workers) in the Lucas (1978) framework. When  $0 < t_m < 1$  entrepreneurs supply *both* managerial and labor inputs to their own businesses. Specifically, entrepreneurs supply labor input equal to the product of their working ability ( $z_w$ ) and the time devoted to non-managerial activities ( $1 - t_m$ ). The total labor input used by an entrepreneur is the sum of the labor supplied by the entrepreneur ( $(1 - t_m)z_w$ ) and the labor hired in the market ( $n_d$ ) from workers outside the family,  $n = n_d + (1 - t_m)z_w$  where  $z_w$  is the working ability of the household. All entrepreneurs choose  $t_m > 0$  because the managerial input is needed for production. Entrepreneurs can be partitioned into two subgroups depending on whether they hire outside labor or not. The first subgroup is given by the employers, who are those entrepreneurs hiring labor outside the family ( $n_d > 0$ ). Entrepreneurs that hire outside labor are assumed to incur a fixed per period operating cost of  $c_e$ .<sup>4</sup> The second subgroup are the self-employed, comprising entrepreneurs that only use their own household labor input ( $n = (1 - t_m)z_w$  and  $n_d = 0$ ). Workers are those households who use all their time as workers ( $t_m = 0$ ), obtaining labor earnings  $wz_w$ .

<sup>2</sup>See, for instance, Erosa (2001), Erosa and Hidalgo-Cabrillana (2008), Jeong and Townsend (2007), Buera and Shin (2011), Buera et al. (2011), Greenwood et al. (2013), Midrigan and Xu (2014) and Moll (2014). Buera et al. (2015) provide a recent and comprehensive survey of this literature. Also, see Quadrini (2009) for a survey of papers on entrepreneurship.

<sup>3</sup>Gollin (2008) emphasizes the dominant presence of small businesses in developing economies. Hurst and Pugsley (2011) and Yurdagul (2017) study, respectively, the role of non-pecuniary benefits of entrepreneurship and a preference for flexible work hours in explaining the low entrepreneurial returns. Poschke (2013) emphasizes the option value of entrepreneurship.

<sup>4</sup>The fixed cost is introduced so that employers demand a non-trivial amount of labor (an amount bounded away from zero), thereby making the distinction between self-employed and employer meaningful.

## 2.2. Stochastic process on skills

Household's skills evolve stochastically over time and there are no insurance markets to insure skill risk. The logarithm of skills of household  $i$  is governed by

$$\ln(z_{wit}) = \alpha_{wi} + u_{wit}, \quad (2)$$

$$\ln(z_{mit}) = \alpha_{mi} + u_{mit} + \zeta_{it}, \quad (3)$$

where  $z_{wit}$  ( $z_{mit}$ ) denotes the working (managerial) skills at date  $t$ ,  $\alpha_{wi}$  and  $\alpha_{mi}$  represent household fixed effects on working and managerial productivities,  $\zeta_{it}$  is a transitory shock to managerial ability, and  $u_{wit}$  and  $u_{mit}$  denote persistent shocks to skills. The latter shocks follow a first order auto-regressive process

$$u_{jit} = \rho_j u_{jit-1} + \epsilon_{jit}, \text{ for } j = w, m, \quad (4)$$

with  $\epsilon_t = (\epsilon_{wt}, \epsilon_{mt})$  jointly drawn from a bivariate normal distribution with correlation coefficient  $\text{corr}(\epsilon_{wt}, \epsilon_{mt}) = \rho_{\epsilon_w, m}$ .

The transitory shock  $\zeta_{it}$  on managerial ability is drawn every period from a normal distribution with mean zero and variance  $\sigma_\zeta^2$ . We assume that each period there is a (small) probability that households draw new skills from the invariant distribution of abilities. In particular, with probability  $p_\alpha$  the household draws a new pair of fixed effects  $(\alpha'_w, \alpha'_m)$  from a fixed bivariate distribution  $F_{(\alpha_w, \alpha_m)}$ .

## 2.3. Capital markets and occupational choice

The financial intermediation industry is assumed to be competitive. Intermediaries take deposits from households and pay the interest rate  $r$ . They rent capital to entrepreneurs at a rate  $r + \delta$ . Enforcement problems limit the capital rented to entrepreneurs. Following Buera and Shin (2013) and Moll (2014), among others, the capital rented by entrepreneurs is limited by a collateral constraint  $k \leq \lambda a$ , where  $a$  denotes household assets. The parameter  $\lambda \geq 1$  determines the degree of financial frictions, with  $\lambda = \infty$  indicating perfect financial markets and  $\lambda = 1$  an economy with no credit.

Given steady state factor prices  $(w, r)$ , the profit made by an entrepreneur in state  $(a, z_w, z_m)$  is given by the function

$$\pi(a, z_m, z_w) \equiv \max_{k, n, 0 \leq n_d \leq t_m \leq 1} \{m^\gamma k^\eta n^\theta - wn_d - rk - \delta k - c_e \mathbf{I}_{n_d > 0}\} \quad (5)$$

s.t.  $k \leq \lambda a$ ,  $m = t_m z_m$ ,  $n = (1 - t_m)z_w + n^d$ ,

The household problem is given by

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} E \left\{ \sum_{t=0}^{\infty} \beta^t \ln(c_t) \right\} \quad (6)$$

s.t.  $c_t + a_{t+1} = \max\{wz_w, \pi(a, z_m, z_w)\} + (1 + r)a_t$ ,  $a_0$  given,

where the max in the budget constraint represents the occupational choice decision.

## 3. Time Allocation and Occupational Maps

We now study, in partial equilibrium (e.g. for a fixed wage rate and interest rate), how the theory can give rise to three active occupational choices: worker, self-employed, and employer. When capital markets are perfect, occupational choices are entirely determined by the ability ratio  $\frac{z_w}{z_m}$ . Individuals with a high  $\frac{z_w}{z_m}$  ratio have a comparative advantage at working and choose to become workers, individuals with a low  $\frac{z_w}{z_m}$  ratio have a comparative advantage at entrepreneurship and choose to become employers, and those with intermediate skill ratios prefer to be self-employed. The discussion below also characterizes how tight borrowing constraints distort occupational choices and the rates of return to skills (see Appendix A for the analytical derivations).

Consider the determinants of self-employment income. Self-employed individuals choose how much time to allocate to managerial versus working activities and how much capital to use in production. The marginal product of their time is equated across its two uses (managerial and working time) and satisfies:

$$MPT_{se} = r_{mw}(\mu) (z_m^\gamma z_w^\theta)^{\frac{1}{\gamma+\theta}}, \text{ where } r_{mw}(\mu) = \left[ \frac{\gamma^\gamma \eta^\eta \theta^\theta}{(r + \delta + \mu)^\eta} \right]^{\frac{1}{1-\eta}} \quad (7)$$

is the rate of return to the composite skill input  $(z_m^\gamma z_w^\theta)^{\frac{1}{\gamma+\theta}}$  and  $\mu$  is the Lagrange multiplier associated with the borrowing constraint.<sup>5</sup> The marginal product of the self-employment time is the product of the skill composite  $(z_m^\gamma z_w^\theta)^{\frac{1}{\gamma+\theta}}$  and the rate of return  $r_{mw}$ . Note that the return to the skill composite decreases with  $\mu$ . The income of a self-employed individual with assets  $a$  is given by

$$y_{se} = r_{mw}(\mu) (z_m^\gamma z_w^\theta)^{\frac{1}{\gamma+\theta}} + \mu k + r a, \quad (8)$$

where  $k = \lambda a$  and  $\mu$  vary across the self-employed with different characteristics  $(z_m, z_w, a)$ .

Now consider the decisions of employers. The marginal product of employers' time ( $MPT_e$ ) can be expressed as the product of managerial skills  $z_m$  and the rate of return  $r_m(\mu)$  on the employer's managerial skill:

$$MPT_e = z_m r_m(\mu) \geq z_w w \text{ with strict inequality if } t_m = 1, \text{ where}$$

$$r_m(\mu) = \gamma \left[ \left( \frac{\eta}{(r + \delta + \mu)} \right)^\eta \left( \frac{\theta}{w} \right)^\theta \right]^{\frac{1}{1-(\eta+\theta)}} \quad (9)$$

Note that  $r_m(\mu)$  is the rate of return to the managerial input  $z_m$  and  $\mu$  is the Lagrange multiplier associated to the borrowing constraint.<sup>6</sup> Using the linear homogeneity of the production function, the income of an employer with ability  $(z_m, z_w)$  and assets  $a$  can be written as

$$y_e = z_m r_m(\mu) + \mu k + r a - c_e, \quad (10)$$

where  $k = \lambda a$  (see Appendix A). Note that borrowing constraints ( $\mu$ ) generate heterogeneity in rates of return to skills among employers. The results below characterize the occupational choice decisions in an economy with perfect enforcement.

**Proposition 1: Perfect Enforcement.** Assume that  $\lambda = \infty$ . Let  $R_1^* \equiv \left( \frac{r_{mw}^*}{w} \right)^{\frac{\theta+\gamma}{\gamma}}$  and  $R_2^* \equiv \left( \frac{r_m^*}{r_{mw}^*} \right)^{\frac{\theta+\gamma}{\theta}}$ , where  $r_{mw}^*$  and  $r_m^*$  are the rates of return to the skill composite  $(z_m^\gamma z_w^\theta)^{\frac{1}{\gamma+\theta}}$  and the managerial skill defined in (7) and (9) when  $\mu = 0$ . i) The rates of return on skills do not vary across individuals ( $r_m = r_m^*, r_{mw} = r_{mw}^*$ ) so that income inequality is all due to heterogeneity in skills and assets. ii) If there is no fixed cost of operation ( $c_e = 0$ ), then the optimal occupational choice is the one that maximizes the marginal product of time and is only determined by the skill ratio  $(\frac{z_w}{z_m})$  as follows: Individuals with an ability ratio  $\frac{z_w}{z_m} > R_1^*$  work for a wage, individuals with  $R_1^* > \frac{z_w}{z_m} > R_2^*$  are self-employed, and those with  $\frac{z_w}{z_m} < R_2^*$  are employers. iii) If the fixed cost of operation is positive ( $c_e > 0$ ), the decision to be an employer depends on the skill ratio  $(\frac{z_w}{z_m})$  and on the absolute level of managerial ability ( $z_m$ ). Individuals prefer to become employers relative to self-employment when  $\frac{z_w}{z_m} < R_2^* \left( 1 - \frac{c_e}{z_m r_{mw}^*} \right)$ .

When there is perfect enforcement (and the fixed cost of operation is zero), occupational choices can be characterized in terms of two skill ratios  $R_1^*$  and  $R_2^*$ . Individuals with comparative advantage at working (an

<sup>5</sup>Although the tightness of the borrowing constraint  $\mu$  is a function of asset holdings  $a$  and skills  $(z_m, z_w)$ , we do not explicitly reflect this in our notation (e.g. write  $\mu(a, z_m, z_w)$ ) to simplify the exposition.

<sup>6</sup>In the calibrated economy, 90% of employers choose  $t_m = 1$ , and the difference in aggregate managerial input ( $t_m * z_m$ ) and aggregate managerial ability ( $z_m$ ) of employers is 1%.

ability ratio  $\frac{z_w}{z_m} > R_1^*$ ) work for a wage, individuals with  $R_1^* > \frac{z_w}{z_m} > R_2^*$  are self-employed, and those with comparative advantage at managing ( $\frac{z_w}{z_m} < R_2^*$ ) are employers. A positive fixed cost of operation ( $c_e > 0$ ), implies that employers require a minimum scale in order to operate a profitable business so that the decision to be an employer depends both on the skill ratio  $\frac{z_w}{z_m}$  and on the level of managerial ability  $z_m$ . Proposition 2 characterizes occupational choice decisions when borrowing constraints bind ( $\lambda$  finite).

**Proposition 2: Imperfect Capital Markets** *Assume that  $\lambda < \infty$  and consider individuals for which the borrowing constraint binds ( $\mu > 0$ ). i) The rates of return on skills ( $r_m, r_{mw}$ ) vary across individuals and decrease with  $\mu$ . Skill returns positively covary with asset holdings. Income inequality is due to heterogeneity in skills, assets, and rates of return. ii) Occupational choices depend on the skill ratio, assets, and the absolute level of skills (the last two affect the rates of return to skills because they determine the tightness of the borrowing constraint).*

Capital market imperfections distort returns to skills and, thus, occupational choices. A tight borrowing constraint depresses the rate of return to the managerial ability of employers and the return to the composite skill input supplied by self-employed individuals. It also increases the rate of return to capital faced by entrepreneurs. For fixed skills ( $z_m, z_w$ ), an increase in assets makes the borrowing constraint less tight ( $\mu$ ), increasing the rates of return to skills and depressing the shadow cost of capital. As a result, asset holdings matter importantly for occupational choice decisions in the presence of financial frictions. Panel a in Figure 3 shows how occupational choices vary with skills in our calibrated model economy for an individual with assets equal to the median in the economy. Panel b draws the occupational map when perfect enforcement is introduced in the calibrated model economy. A comparison of the occupational maps, reveals that capital market imperfections expand the region where self-employment is optimal at the expense of the regions where employer and worker are the preferred occupational choices.

#### 4. Calibration

The baseline model economy is calibrated to assess the quantitative implications of the theory. To benchmark the results, a version of the model economy in which entrepreneurs use all their time to manage their businesses and households are not heterogeneous in working skills is also calibrated. This version of the model economy is labelled as the “standard model economy”.

To simplify the calibration, some parameters are set (externally) using estimates from other studies in the literature. In particular, for the two calibrated model economies, the parameters of the production function are set to standard values in the literature:  $\gamma = 0.198, \eta = 0.3256, \theta = 0.4764$  (see Guner et al. (2008)). The model period is set to a year and the annual depreciation rate of capital is set to  $\delta = 0.06$ . The rest of the parameters are assigned by solving the model economy and minimizing a loss function. The loss function consists of the sum of square deviations between key moments in the model economy and in the Brazilian data.

The household level data used to calibrate the model comes mainly from the Monthly Labor Survey (PME for its name in Portuguese, *Pesquisa Mensal de Emprego*) for the periods 2003-2010. The PME has a sampling structure similar to the CPS in the US: selected households are interviewed for four months, stay out of the sample for the next eight and then they are again interviewed for four months before leaving the sample. This structure allows us to exploit the short panel dimension to compute transitions of earnings and occupations. The calibration also targets consumption data from the Household Expenditure Survey (POF for its name in Portuguese *Pesquisa de Orçamento Familiar*) for the period 2008/2009.<sup>7</sup>

##### 4.1. Calibration of the baseline economy

The managerial fixed effects ( $\alpha_m$ ) are assumed to be drawn from a standard Pareto distribution with tail parameter  $\eta_m$ . The location of the Pareto distribution is shifted with a parameter  $\mu_m$  that controls the mean value of managerial fixed effects. The Pareto distribution is discretized and the first grid point is set to the one that accumulates probability mass  $p_m$ . The fixed effects on working ability are assumed to be

<sup>7</sup>See Appendix C for further details about the datasets.



drawn from a Normal distribution with variance  $\sigma_{wf}^2$  and a mean value that depends on the realization of the managerial fixed effect according to a parameter  $\rho_{\alpha_{mw}}$  that controls the correlation of fixed effects on working and managerial skills.<sup>8</sup>

In total 13 parameters are calibrated internally by minimizing the distance between moments in the model and 15 moments in the data. While the moments targeted are all jointly determined by all of these parameters, in presenting the calibration procedure it is useful to associate to each target a specific parameter.

The calibration targets the following moments in the Brazilian data: First, the mean fixed effect on managerial ability ( $\mu_m$ ) and the fixed cost of employers ( $c_e$ ) to match the occupational choice structure with 68% of workers, 24% of self-employed, and 8% of employers (2 independent targets). Second, the variance of shocks to working ability ( $\sigma_{\epsilon_w}^2$ ) and the Pareto tail on fixed effects on managerial ability ( $\eta_m$ ), to match the variance of log-earnings of workers and entrepreneurs with values of 0.76 and 1.06 (2 targets). Third, the variance and serial correlation of innovation to managerial skills ( $\sigma_{\epsilon_m}^2, \rho_m$ ), the correlation between innovations to working and managerial skills ( $\rho_{\epsilon_m, w}$ ) and the transitory shock to managerial ability ( $\sigma_{\zeta}^2$ ) to match the yearly transition rates across occupations (worker to worker, worker to employer, self-employed to self-employed, self-employed to employer, employer to self-employed, employer to employer) (6 independent targets). Fourth, the correlation of fixed effects on working and managerial skills ( $\rho_{\alpha_{m, w}}$ ) and the probability of the lowest fixed effect on managerial skill ( $p_m$ ) to match the difference in mean log-earnings between employers and workers (0.66), between employers and self-employed (0.94), and between workers and entrepreneurs (0.04) (2 independent targets). Fifth, the probability of drawing new fixed effects ( $p_\alpha$ ) to match the variance of log consumption of 0.80 (1 target). Sixth, the parameter on the collateral constraint ( $\lambda$ ) to match the credit to (annual) GDP ratio of 42% (1 target).<sup>9</sup> Seventh, the discount factor ( $\beta$ ) to match the capital to output ratio of 2.1 (1 target).<sup>10</sup>

#### 4.2. Calibration of standard model economy

Following Buera and Shin (2011), the shocks to managerial skill are specified to follow an AR(1) process. The calibration involves choosing four parameters to match four moments in the data: (i) the serial correlation of managerial shocks to match the persistence of entrepreneurship between two consecutive years of 87%; (ii) the variance of innovations to managerial shocks to match the variance of log entrepreneurial income of 1.06; (iii-iv)  $\lambda$  and  $\beta$  to match the target for the credit to GDP ratio (42%) and the capital to income ratio in the Brazilian data (2.1).

#### 4.3. Calibration results

The parameter values and the calibration results for the baseline economy are reported in Panel A of Tables 1 and 2. Panel B in these tables reports similar statistics for the standard model economy. While the two calibrated model economies match the calibration targets well, the baseline economy provides a much better account of the occupational structure, income differences *between* and *within* occupations, and household inequality in Brazil.

The baseline model economy matches perfectly the occupational structure in Brazil: It matches the overall entrepreneurship rate in Brazil of 32% and the composition of entrepreneurs between self-employed household (24%) and employers (8%). These observations stand in contrast to the standard model economy, which implies an entrepreneurship rate less than a third the one in the data (10%) and cannot account for self-employment. The baseline model can match the fraction of self-employed households in the data with

<sup>8</sup>We follow the calibration strategy in Storesletten et al. (2005) to pin down  $\rho_w$  and  $\sigma_{wf}^2$ . Appendix B presents a detailed description of the discretization of the skill distribution as well as a discussion of the calibration strategy. It also reports on the performance of the model economy on non-targeted dimensions.

<sup>9</sup>See The World Bank (2017).

<sup>10</sup>The capital to output ratio was computed following the same methodology as in Greenwood et al. (2013) using the Penn World Tables 7.0 (see Heston et al. (2011)), and corresponds to the average for the year 2004-2010.

modest fixed cost of operation by employers. The fixed cost of operation faced by employers in the baseline economy represents 0.9% of GDP.<sup>11</sup>

The baseline model quantitatively accounts for the substantial heterogeneity in earnings *within* and *between* occupations. The variance of log-earnings among workers and entrepreneurs are 0.76 and 1.07 in the baseline economy (versus 0.76 and 1.06 in the data). The theory is also consistent with observed earnings inequality between occupations. Mean earnings are higher for employers than for workers, and are the lowest for self-employed households. The difference in mean log-earnings between employers and workers is 0.66 (0.66 in the data) and between employers and self-employed is 0.93 (0.94 in the data). Self-employed households make on average 28% less earnings than workers (29% in the data). To match these facts on earnings differences across occupations, the theory requires a positive but moderate correlation of managerial and working skills (the correlation of fixed effects is 0.145).

By design, the standard model economy cannot account for the substantial earnings heterogeneity across workers in the data because it assumes away heterogeneity in working ability. Moreover, because it cannot distinguish between comparative advantage and absolute advantage at entrepreneurship, it largely overstates earnings differences between occupations: The model implies a difference in mean log-earnings between entrepreneurs and workers of 1.35 versus the value of -0.04 in the data. On the other hand, the model generates too little inequality across households since the variance of log consumption is 0.32 relative to 0.80 in the data.

The baseline economy is consistent with key data patterns on occupational transitions, though the match is not perfect. First, the model predicts that being a worker is quite persistent: 94% of workers in the model economy and in the data are workers one year later. Second, both in the baseline economy and in the data, entrepreneurs are less likely to remain in their occupations than workers. The model matches pretty closely the persistence of self-employment in the data (75% versus 77%) but overstates the persistence of employers (76% vs 68%). Third, the baseline model economy matches the fact that households are much more likely to transit into employer from self-employment than from being a paid worker. The flow of self-employed households into employers is 8% in the data and 7% in the model. The annual transition rate from worker into employer is quite low both in the model and in the data (1% in the data and negligible in the model). Fourth, the model is consistent with the fact that employers are much more likely to switch to self-employment than paid work.

Summing up, the calibrated model economy with heterogeneity in two skills provides a reasonable account of occupational choices, income inequality within and across occupations, occupation transitions, and household inequality in Brazil. In Appendix B, we evaluate the model economy along many non-targeted dimensions.<sup>12</sup>

## 5. Financial frictions, entrepreneurship, and the cross-country data

The quantitative performance of the theory is assessed by computing, for the two calibrated model economies, the equilibrium allocations for economies that differ in the parameter  $\lambda$  determining the strength of financial frictions.<sup>13</sup>

Figure 4a plots the relationship between the entrepreneurship rate and the ratio of external finance to GDP in the model-simulated data and in the cross-country data. The short-dashed line corresponds to the

<sup>11</sup> Alternatively, the fixed cost represents 0.7% of the mean output of employers or 18% of the mean wage in the economy. Buera et al. (2011) calibrate a fixed cost of operation for firms in the manufacturing sector that is about 3 times the average wage. In Midrigan and Xu (2014) the fixed costs of operation represent 4.6% of GDP, which is about 5 times larger than the one we obtain.

<sup>12</sup> In particular, the calibrated model economy is consistent with (i) the fact that within group (occupation) inequality explains 96% of the variation of earnings in the data; (ii) non-targeted moments of the earnings distribution; (iii) salient features regarding mean earnings gains/losses for workers/employers/self-employed households switching to other occupations versus those that do not switch; (iv) measures of consumption inequality as well as differences in mean consumption across households in different occupations.

<sup>13</sup> Equilibrium is computed for 19 economies with  $\lambda$  varying in the interval (1,2500). The grid for  $\lambda$  is not equally spaced since the grid size is smaller when  $\lambda$  is close to 1. Equilibrium statistics do not change for values of  $\lambda > 2500$ . The economy with  $\lambda = 2500$  is interpreted as an economy with perfect credit markets.

data generated from the baseline economy, the long-dashed line from the standard model economy, and the solid line to the actual cross-country data. Relative to the data, the rate of entrepreneurship predicted by the standard model economy is too low and varies too little with the level of financial development. The range of variation of the entrepreneurship rate in this model goes from 7% to 11.5%. The baseline model economy accounts much better for the relationship between entrepreneurship and financial development in the cross-country data. The range of variation of entrepreneurship predicted by this model economy goes from 12% to 33% (which is about 5 times the one predicted by the standard model). To gain some quantitative sense on the performance of the model economies relative to the data, the rate of entrepreneurship is regressed on external finance to GDP in the cross-country and model simulated data. We find that the regression coefficient of external finance is -0.11 in the baseline economy, -0.025 in the standard model, and -0.23 in the data. The baseline economy accounts for half of the variation in the data which is quite impressive giving that only one factor (financial frictions) is varying across economies. The standard model on the other hand explains only about one tenth of the response of entrepreneurship to changes in external finance to GDP.

In the baseline economy the fractions of employers and self-employed households vary in opposite directions with financial development. While the rate of self-employed households is strongly decreasing with the level of financial development, the fraction of employers shows a mildly positive relationship with external finance to GDP ratio (see Figures 4b and 4c).<sup>14</sup> Both of these implications are validated by the data: first, the regression coefficient of the fraction of employers on external finance to GDP is 0.01 both in the model and in the data. Second, the regression coefficient of the fraction of self-employed households on credit to GDP ratio is -0.12 in the model and -0.24 in the data.

In the baseline economy, a reduction in financial frictions shifts households from self-employment into paid work, thereby increasing the share of employee compensation in national income. To test this prediction of the theory, Figure 5 graphs the relationship between the ratio of employee compensation to GDP and the ratio of external finance to GDP in the model-simulated data and in the cross-country data. The baseline model economy is consistent with the fact that these two variables are positively associated in the cross-country data. The standard model cannot account for this evidence because it abstracts from self-employed households. Quantitatively, the baseline economy accounts for about 30% of the empirical association between the ratio of employee compensation to GDP and the ratio of external finance to GDP.

Consider the effects of financial frictions in output per worker and TFP.<sup>15</sup> The coefficient of the regression of output per worker on credit to GDP is 0.23 for the baseline model, 0.24 for the standard model, and 0.34 for the data. In a similar exercise Buera et al. (2011) find a regression coefficient of 0.22 in their model economy. Hence, consistently with their analysis, we conclude that the model accounts for about two thirds of the cross-country relationship between output per worker and financial development. The regression coefficient of TFP on the external finance to GDP ratio is 0.12 for both the baseline and the simple model, and 0.26 for the cross-country data. Hence, the two calibrated model economies account for about 50% of the cross-country association between TFP and financial development. Nonetheless, the next section of the paper shows that modeling heterogeneity in two skills amplifies substantially the estimates of the gains in output per worker and TFP of removing financial frictions in Brazil. It also amplifies the effect of financial frictions on household inequality.

## 6. The impact of removing financial frictions in Brazil

This section studies in detail the impact on entrepreneurship, resource allocation and production efficiency of removing financial frictions in Brazil. This is done by comparing some key model statistics in the baseline economy with those of an identical economy with perfect credit markets.

<sup>14</sup>The model economy was calibrated to Brazil. Since the fraction of employers in Brazil is well above the implied by the regression line in the cross-country data, the regression line for the model is above the one in the data.

<sup>15</sup>TFP is measured as in standard growth accounting exercises  $TFP = \frac{Y}{K^{\frac{\eta}{\eta+\theta}}}$ , where Y denotes aggregate output (net of fixed costs) and K aggregate capital.

### 6.1. Output, credit, and entrepreneurship

The elimination of financial frictions leads to a much larger increase in output in the baseline economy than in the standard model economy (58% versus 47%, as shown in columns a and b in Table 3). In understanding this result, it is useful to decompose the impact of financial frictions in output in two terms: how much output changes in response to an increase in the credit to GDP ratio and how much the credit to GDP ratio changes in response to a decrease in financial frictions. The first effect was estimated in the previous section of the paper where it was found that the two calibrated model economies imply similar regression coefficients on credit to GDP on output. However, the second effect is much stronger in the baseline model economy: The elimination of financial frictions leads to an increase in the credit to GDP ratio of 1.53 in the baseline economy and of 1.28 in the standard model economy, explaining why the increase in aggregate output is larger in the former economy.

The key to this result is that the decrease in the fraction of entrepreneurs is more than 5 times bigger in the baseline economy than in the standard economy (0.20 versus 0.035). The larger reduction in the number of entrepreneurs in the baseline economy is associated with a larger increase in the average productivity of entrepreneurs. While the mean skill of entrepreneurs increases by about 200% in the baseline economy, it increases by about 50% in the standard economy. The reallocation of capital and labor towards the most productive entrepreneurs is thus stronger in the baseline economy which, in turn, accounts for the larger credit expansion in this economy.

### 6.2. Resource allocation and TFP

Using the production function of the model economy, the efficiency of production is measured as  $TFP_{ideal} = Y/K^\eta N^\theta$ , where  $Y$  denotes the aggregate output (net of fixed costs) and  $(K, N)$  the aggregate capital and labor input. The gains in production efficiency are much larger in the baseline economy than in the standard model economy (26% versus 21%) despite the fact that the two model economies imply the same regression coefficient of external finance relative to GDP on  $TFP_{ideal}$  (0.15). Again, this finding is explained by the larger credit expansion in the baseline economy. We also measure TFP as in standard growth accounting exercises:  $TFP_{data} = Y/K^{\frac{\eta}{\theta+\eta}}$ . The changes in measured TFP are smaller than the ones in  $TFP_{ideal}$  for both economies (23% in the baseline economy versus 19% in the standard economy). Hence, standard TFP measurement underestimates the true production efficiency gains by 3 percentage points in our baseline economy and by 2 percentage points in the standard economy.

The baseline economy has novel implications on how financial frictions affect output and production efficiency by employers and self-employed entrepreneurs. In the baseline economy, the elimination of financial frictions changes dramatically the contribution to aggregate output from employers and self-employed entrepreneurs. While the aggregate output of employers increases by 80 percent, the aggregate output of the self-employed decreases by 87 percent. As a result, the share of aggregate output produced by self-employed households decreases from 13 percent in the baseline economy to 1 percent in the economy with no financial frictions.

The elimination of financial frictions leads to large production efficiency gains by employers (28%) and large losses by self-employed households (-14.3%). Standard growth accounting measures of TFP underestimate the gains by employers and the losses by self-employed entrepreneurs (14% relative to 28% for employers and -11% relative to -14.3% for self-employed). These differences are large because there are large changes in the allocation of capital and employment across the two types of entrepreneurs. The large increase in the capital and employment allocated to employers reduces their measured TFP gains relative to their true productivity gains and the opposite occurs for the self-employed.

### 6.3. Financial frictions and inequality

The elimination of financial frictions leads to a reduction of the Gini coefficient in assets from 0.91 to 0.84, decreases the Gini coefficient of consumption from 0.60 to 0.56, reduces the auto-correlation of earnings from 0.89 to 0.80, and decreases the correlation of earnings and assets from 0.71 to 0.38. These effects are

present in the standard model economy but to a lesser extent. Interestingly, financial frictions have a large effect on the *sources* that drive inequality.<sup>16</sup>

Table 4 decomposes the variance of income that is due to the variance of capital income, returns to skills, and their covariance. Financial frictions diminish the inequality in skill returns by limiting the rents that talented individuals obtain from their skills. Since entrepreneurs with high managerial skills need to operate at a large scale to obtain all the potential rents from their skills, borrowing constraints diminish the rate of return attained by these individuals (see Proposition 2). The elimination of financial frictions allows talented individuals to obtain the full reward to their skills.

The variation in skill returns explains 20% of the income variance in the baseline economy whereas it accounts for 64% of the income variance in the economy with no financial frictions (see Table 4). Capital income (directly) accounts for 31% of the income variance in the baseline economy, which is about twice the 12% figured in the PCM economy.

#### 6.4. Sensitivity analysis

The sensitivity of the results to changes in some key parameters and modeling choices is now briefly examined. In a first experiment, the variance of working skills in the baseline economy is set to zero. This experiment allows to isolate how modeling the entrepreneurial time allocation decision matters for the quantitative impact of financial frictions vis a vis the effect of heterogeneity in working skills. The baseline economy assumes that employers pay a fixed cost of operation. In a second experiment, it is examined how the results vary when this cost is set to zero or when it is expressed in terms of the wage rate (so that an increase in the equilibrium wage rate rises the cost of operation). The results are presented in columns c to e in Table 3.<sup>17</sup>

When financial frictions are eliminated in the economy with no heterogeneity in working skills, the decrease in the fraction of self-employed workers is 0.09 which is about a half the one in the baseline economy (0.20). Hence, modeling the entrepreneurial time allocation decision and heterogeneity in working ability are both important for the quantitative response of entrepreneurship to financial frictions.

The large credit expansion (associated to the large reduction of the entrepreneurship rate) explains why aggregate output and TFP increase more with the removal of financial frictions in the baseline economy than in the standard model economy. Table 3 offers some interesting clues for understanding this result. Financial frictions affect labor demand and labor supply through changes in the skill distribution of workers and entrepreneurs. The elimination of financial frictions increases the average managerial quality of entrepreneurs augmenting aggregate labor demand. This effect is much weaker in the standard model economy than in the other economies (the average skill of entrepreneurs rises by about 50% in the standard economy and by more than 180% in the other economies). The economies with heterogeneity in working skills feature an additional effect: The removal of financial frictions leads to a decrease in the average ability of workers of more than 8%, reducing the effective aggregate labor supply. Hence, a large change in the fraction (quantity) of workers is needed to equate labor supply and labor demand in these economies.

The calibrated fixed cost is not important for the output gains of removing financial frictions: The output gain is 55% in the economy with no fixed costs ( $c_e = 0$ ), 57% when the fixed cost rises with the wage rate, and 58% in the baseline economy (columns a, d, e in Table 3). While  $c_e$  has a small effect on how financial frictions affect the entrepreneurship rate, it matters for the effect of financial frictions on the *composition* of entrepreneurs. The fraction of employers does not change with the elimination of financial frictions in the baseline economy. However, when  $c_e = 0$  the decrease in entrepreneurship is explained by a substantial reduction in both the fractions of self-employed (12%) and employers (10%), which is at odds with the cross-country data.

<sup>16</sup>In the standard model, the elimination of financial frictions diminishes the Gini coefficient of assets from 0.93 to 0.89, the Gini of consumption from 0.386 to 0.364, and the correlation of earnings and assets from 0.71 to 0.41. To measure the sources of inequality, returns to skills are computed as the total rents on working and managerial skills:  $wz_w$  for workers,  $r_m z_m$  for employers, and  $r_{mw} z_{mw}$  for self-employed, as defined in Section 3.

<sup>17</sup>Following a suggestion from an anonymous referee, Appendix B.4 shows how these model economies fit the calibration targets of the baseline model economy.

## 7. Conclusions

The Lucas (1978) model is extended to distinguish between entrepreneurs that are self-employed from those that are employers. The proposed theory assumes that households are heterogeneous in managerial and working skills and entrepreneurs decide how much time to devote to the supply of labor and/or managerial input. While the baseline model accounts for half of the association in the data between entrepreneurship and external finance to GDP, the standard model explains only about one tenth of the cross-country variation in entrepreneurship. The baseline model is also consistent with the fact that the fraction of employers shows a mildly positive association with the external finance to GDP ratio and that the cross-country variation in entrepreneurship is mostly explained by the variation in self-employment rates. Moreover, the output gain of eliminating financial frictions in the baseline economy is much larger than in the standard model (58% versus 47%). The larger reallocation of capital towards a smaller number of entrepreneurs in the baseline economy implies a larger credit expansion and output gain.

## References

### References

- Buera, F. J., Kaboski, J. P., Shin, Y., August 2011. Finance and development: A tale of two sectors. *American Economic Review* 101 (5), 1964–2002.
- Buera, F. J., Kaboski, J. P., Shin, Y., 2015. Entrepreneurship and financial frictions: A macrodevelopment perspective. *Annual Review of Economics* 7 (1), 409–436.
- Buera, F. J., Shin, Y., 2011. Self-insurance vs. self-financing: A welfare analysis of the persistence of shocks. *Journal of Economic Theory* 146 (3), 845–862.
- Buera, F. J., Shin, Y., 2013. Financial frictions and the persistence of history: A quantitative exploration. *Journal of Political Economy* 121 (2), 221–272.
- D'Erasmus, P. N., Boedo, H. J. M., 2012. Financial structure, informality and development. *Journal of Monetary Economics* 59 (3), 286–302.
- Erosa, A., 2001. Financial intermediation and occupational choice in development. *Review of Economic Dynamics* 4 (2), 303–334.
- Erosa, A., Hidalgo-Cabrillana, A., 2008. On finance as a theory of tfp, cross-industry productivity differences, and economic rents. *International Economic Review* 49 (2), 437–473.
- Gollin, D., 2008. Nobody's business but my own: Self-employment and small enterprise in economic development. *Journal of Monetary Economics* 55 (2), 219–233.
- Greenwood, J., Sanchez, J. M., Wang, C., 2013. Quantifying the impact of financial development on economic development. *Review of Economic Dynamics* 16 (1), 194–215.
- Guner, N., Ventura, G., Xu, Y., 2008. Macroeconomic implications of size-dependent policies. *Review of Economic Dynamics* 11 (4), 721–744.
- Hamilton, B. H., 2000. Does entrepreneurship pay? an empirical analysis of the returns to self-employment. *Journal of Political Economy* 108 (3), 604–631.
- Heston, A., Summers, R., Aten, B., 2011. Penn world table version 7.0 center for international comparisons of production. Income and Prices at the University of Pennsylvania.
- Hurst, E., Pugsley, B. W., 2011. What do small businesses do? *Brookings Papers on Economic Activity*, 73–143.
- Jeong, H., Townsend, R. M., 2007. Sources of tfp growth: occupational choice and financial deepening. *Economic Theory* 32 (1), 179–221.
- Jovanovic, B., 1994. Firm formation with heterogeneous management and labor skills. *Small Business Economics* 6 (3), 185–191.
- Lucas, R. E., 1978. On the size distribution of business firms. *The Bell Journal of Economics*, 508–523.
- Midrigan, V., Xu, D. Y., February 2014. Finance and Misallocation: Evidence from Plant-Level Data. *American Economic Review* 104 (2), 422–58.
- Moll, B., 2014. Productivity losses from financial frictions: can self-financing undo capital misallocation? *The American Economic Review* 104 (10), 3186–3221.
- Poschke, M., 2013. Who becomes an entrepreneur? labor market prospects and occupational choice. *Journal of Economic Dynamics and Control* 37 (3), 693–710.
- Quadri, V., 2009. Entrepreneurship in macroeconomics. *Annals of Finance* 5, 295311.
- Storesletten, K., Telmer, C., Yaron, A., 2005. Consumption and risk sharing over the life cycle. *Journal of Monetary Economics* 52 (7), 1331–1358.
- The World Bank, 2017. World development indicators. Data retrieved from World Development Indicators, <http://data.worldbank.org/>.
- Vissing-Jorgensen, A., Moskowitz, T. J., 2002. The returns to entrepreneurial investment: A private equity premium puzzle? *American Economic Review* 92 (4), 745–778.
- Yurdagül, E., 2017. Production complementarities and flexibility in a model of entrepreneurship. *Journal of Monetary Economics* 86, 36–51.

| Panel A.                | Baseline Model Economy                                |        |
|-------------------------|-------------------------------------------------------|--------|
| $p_m$                   | Prob. first fixed effect on manag. skill              | 0.229  |
| $\eta$                  | Pareto tail managerial skill                          | 5.40   |
| $\mu_m$                 | Location managerial skill distrib.                    | -2.976 |
| $\rho_{\alpha_m, w}$    | Correlation working and managerial fixed effects      | 0.145  |
| $\sigma_{\epsilon_w}^2$ | Variance innovation working skill                     | 0.073  |
| $\sigma_{\epsilon_m}^2$ | Variance innovation managerial skill                  | 1.145  |
| $\rho_m$                | Autocorrelation managerial skill                      | 0.788  |
| $\rho_{\epsilon_m, w}$  | Correlation innovations managerial and working skills | 0.335  |
| $\sigma_{\zeta}^2$      | Variance transitory shock managerial skill            | 0.212  |
| $\beta$                 | Discount factor                                       | 0.886  |
| $\lambda$               | Collateral constraint                                 | 1.27   |
| $c_e$                   | Fixed cost of employers                               | 0.08   |
| $p_\alpha$              | Probability of new skills draw                        | 0.041  |
| Panel B.                | Standard Model Economy                                |        |
| $\sigma_m^2$            | Variance innovation managerial skill                  | 1.25   |
| $\rho_m$                | Autocorrelation managerial skill                      | 0.929  |
| $\beta$                 | Discount factor                                       | 0.898  |
| $\lambda$               | Collateral constraint                                 | 1.29   |

Table 1: Calibrated Parameters. *Note:* The *baseline model economy* (Panel A) refers to the economy with heterogeneity in both entrepreneurial and working ability, while in the *standard model economy* (Panel B) there is heterogeneity only in the entrepreneurial ability while workers are homogeneous.

| Panel A: Baseline Model Economy               |       |       |
|-----------------------------------------------|-------|-------|
|                                               | Data  | Model |
| K / Y                                         | 2.11  | 2.04  |
| Credit / Y                                    | 0.42  | 0.43  |
| <i>Income inequality: Within occupations</i>  |       |       |
| Var( $y_w$ )                                  | 0.76  | 0.76  |
| Var( $y_{ent}$ )                              | 1.06  | 1.07  |
| <i>Income inequality: Between occupations</i> |       |       |
| Avg $y_{EMP}$ -Avg $y_w$                      | 0.66  | 0.66  |
| Avg $y_{SE}$ -Avg $y_w$                       | -0.29 | -0.28 |
| Avg $y_{EMP}$ -Avg $y_{SE}$                   | 0.94  | 0.93  |
| <i>Occupational Structure</i>                 |       |       |
| Workers                                       | 68%   | 68%   |
| Self-Employed                                 | 24%   | 24%   |
| Employers                                     | 8%    | 8%    |
| <i>Occupational Transitions</i>               |       |       |
| W to W                                        | 94%   | 94%   |
| W to SE                                       | 5%    | 6%    |
| SE to SE                                      | 77%   | 75%   |
| SE to E                                       | 8%    | 8%    |
| E to SE                                       | 22%   | 23%   |
| E to E                                        | 68%   | 76%   |
| <i>Consumption inequality</i>                 |       |       |
| Var(log(c))                                   | 0.80  | 0.73  |
| Panel B; Standard Model Economy               |       |       |
|                                               | Data  | Model |
| K / Y                                         | 2.11  | 2.0   |
| Credit / Y                                    | 0.42  | 0.42  |
| Persistence Entrep.                           | 0.87  | 0.88  |
| Var( $y_{ent}$ )                              | 1.06  | 1.06  |

Table 2: Calibration Results. *Note:* In the table W stands for Workers, SE for Self-Employed and E for Employers. Variances and differences in earnings between occupations are computed using the natural logarithm of earnings. Occupational transitions are computed comparing the same month of two consecutive years, keeping only households that were employed in both years.

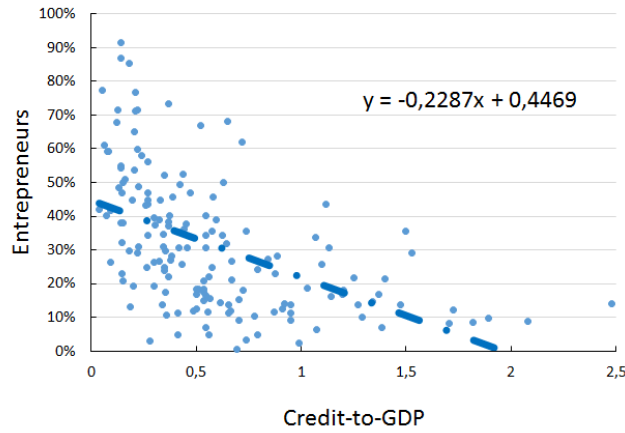


|                              | (a)      | (b)      | (c)              | (d)       | (e)       |
|------------------------------|----------|----------|------------------|-----------|-----------|
|                              | Baseline | Standard | $\sigma_w^2 = 0$ | $c_e = 0$ | $c_e * w$ |
| <i>Change in fraction of</i> |          |          |                  |           |           |
| Workers                      | 0.20     | 0.035    | 0.07             | 0.22      | 0.20      |
| Entrepreneurs                | -0.20    | -0.035   | -0.07            | -0.22     | -0.20     |
| Self-Employed                | -0.20    | 0        | -0.09            | -0.12     | -0.19     |
| Employers                    | 0.0      | -0.035   | 0.03             | -0.10     | -0.01     |
| Credit to GDP                | 1.53     | 1.28     | 1.29             | 1.52      | 1.53      |
| <i>% Change</i>              |          |          |                  |           |           |
| Output                       | 58%      | 47%      | 54%              | 55%       | 57%       |
| Avg skill workers            | -8%      | 0%       | 0%               | -14%      | -8%       |
| Avg manag skill entrep.      | 201%     | 52%      | 180%             | 217%      | 193%      |
| $TFP_{data}$                 | 23%      | 19%      | 26%              | 21%       | 22%       |
| $TFP_{ideal}$                | 26%      | 21%      | 31%              | 24%       | 26%       |

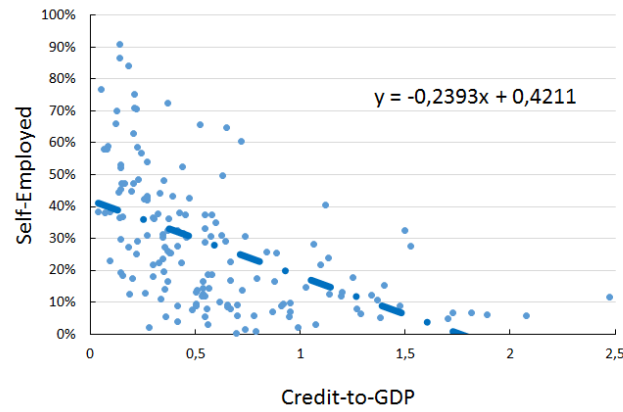
Table 3: Aggregate Effects of Removing Financial Frictions in Brazil. *Note:* This table reports the effect of removing financial frictions in different economies: i) Baseline economy, ii) Standard economy, iii) Baseline economy with variance of working skills set to zero, iv) Baseline economy no fixed costs of operation by employers, iv) Baseline economy with fixed costs in terms of labor input.

|                 | Baseline Model |     | Standard Model |     |
|-----------------|----------------|-----|----------------|-----|
|                 | Baseline       | PCM | Baseline       | PCM |
| Capital income  | 31%            | 12% | 31%            | 17% |
| Skill returns   | 20%            | 64% | 20%            | 57% |
| Covariance term | 49%            | 24% | 48%            | 25% |

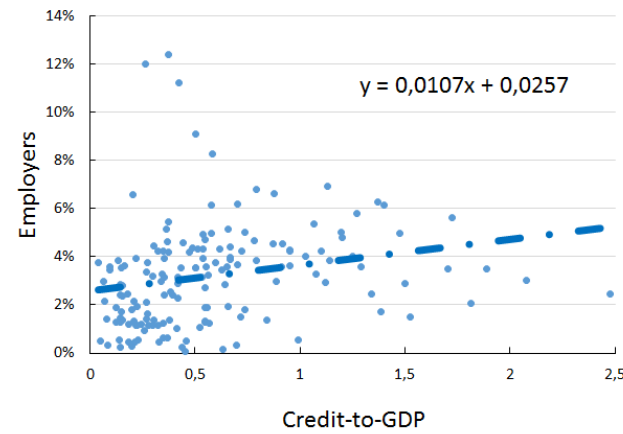
Table 4: Variance Decomposition of Income. *Note:* Total income can be decomposed in capital income and skill returns. Then  $VAR(y) = VAR(\text{skill returns}) + VAR(\text{capital income}) + 2 * COVAR(\text{skill returns}, \text{capital income})$ . PCM refers to the economy with perfect capital markets.



(a) Entrepreneurs



(b) Self-Employed



(c) Employers

Figure 1: Occupational Structure and Credit across Countries. *Note:* Occupational structure data is taken from the International Labor Organization and corresponds to the year 2015. Entrepreneurs are defined as those own-account workers where the remuneration depends on the goods and services produced. Self-employed are entrepreneurs who do not hire labor, while employers are those entrepreneurs who hire at least one person. Credit-to-GDP is taken from the World Bank development indicators database for the year 2015 and is defined as domestic credit to the private sector. All the variables are defined in the same way in the other figures of the paper. The dots indicate countries for which we have data on occupation and credit, while the dashed-line is the linear regression line of occupation on credit.

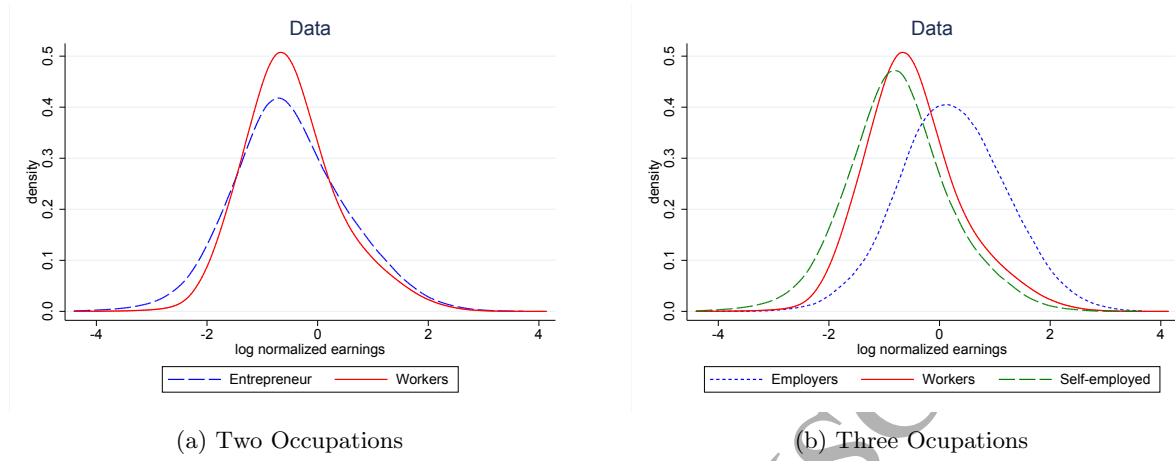


Figure 2: Earnings Distribution in Brazil. *Note:* The figure plots the normalized log-earnings for each occupation. Earnings are effectively perceived earnings by the individual in the month from all the works done. The normalization is done by dividing the earnings of each household by the average earning of the economy. The distributions are estimated Epanechnikov kernel densities, using the STATA command `kdensity` with band width of 0.2.

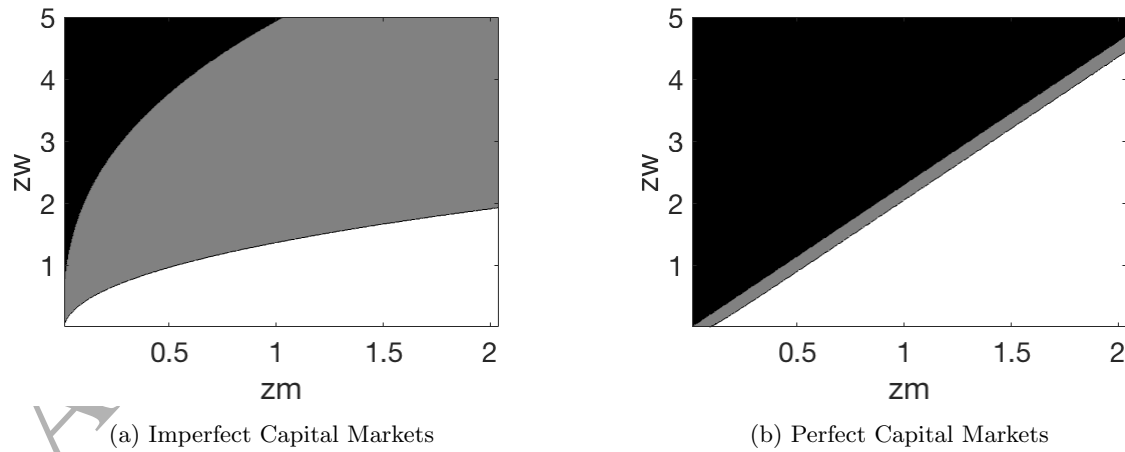
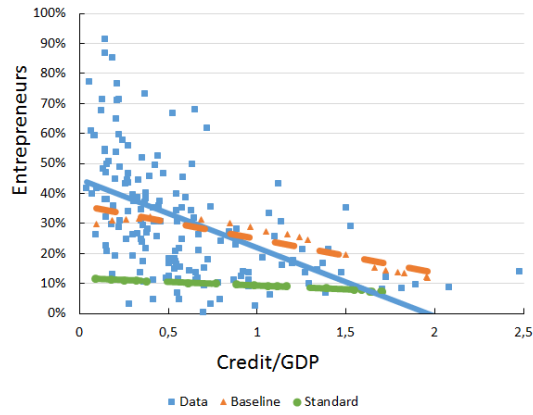
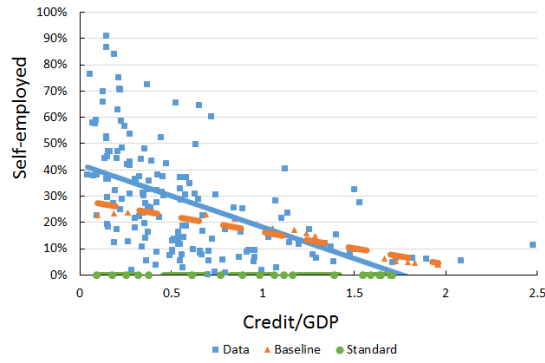


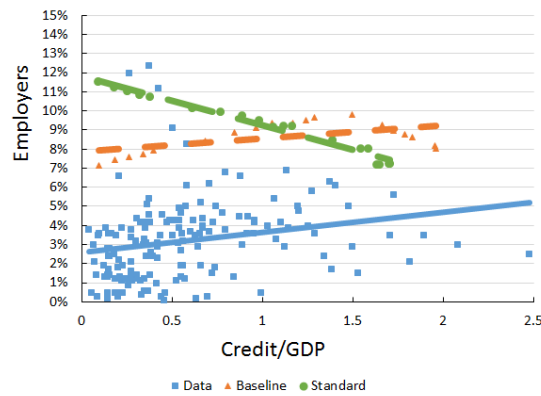
Figure 3: Occupational Map. *Note:* The graphs present the occupational choice for each combination of entrepreneurial and working ability for an individual with the median asset in the baseline economy and for the economy with perfect capital markets (in this case the occupational choices do not depend on the asset level).



(a) Entrepreneurs



(b) Self-Employed



(c) Employers

Figure 4: Occupational Structure and Credit across Countries. *Note:* In the graphs the baseline model corresponds to the economy with heterogeneity in both managerial and working ability, while the standard model corresponds to an economy with heterogeneity only in the managerial ability.

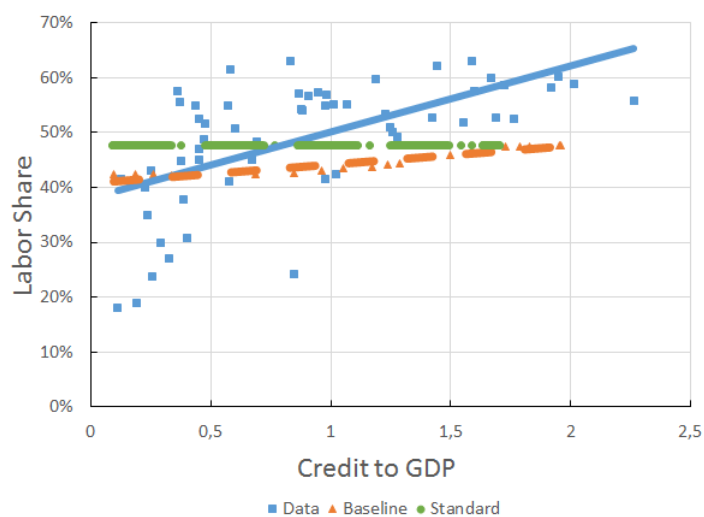


Figure 5: Employee compensation and external finance across countries *Note:* Data for employee compensation is taken from the International Labor Organization, and defined as total remuneration, in cash or in kind, payable by an enterprise to an employee in return for work.